GENERALIZATIONS ARE THE IMAGE OF SPEC OF THE LOCAL RING

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2620-2634, following the statement at 2611-2618.

Allowed standard prerequisites (editorial).

- Canonical morphism Spec(O_{X,x}) -> X. lemma-morphism-from-spec-local-ring (card L26396), canonical morphism constructed at ~2584-2586. Supplies the morphism whose image is characterized.

Lemma 0.1. *Let X be a scheme. Let $x, x' \in X$ be points of X. Then $x' \in X$ is a generalization of x if and only if x' is in the image of the canonical morphism $\mathrm{Spec}(\mathcal{O}_{X,x}) \to X$.*

Proof. A continuous map preserves the relation of specialization/generalization. Since every point of $\mathrm{Spec}(\mathcal{O}_{X,x})$ is a generalization of the closed point we see every point in the image of $\mathrm{Spec}(\mathcal{O}_{X,x}) \to X$ is a generalization of x. Conversely, suppose that x' is a generalization of x. Choose an affine open neighbourhood $U = \mathrm{Spec}(R)$ of x. Then $x' \in U$. Say $\mathfrak{p} \subset R$ and $\mathfrak{p}' \subset R$ are the primes corresponding to x and x'. Since x' is a generalization of x we see that $\mathfrak{p}' \subset \mathfrak{p}$. This means that $\mathfrak{p}'$ is in the image of the morphism $\mathrm{Spec}(\mathcal{O}_{X,x}) = \mathrm{Spec}(R_\mathfrak{p}) \to \mathrm{Spec}(R) = U \subset X$ as desired. □

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